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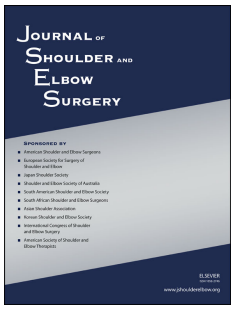


Distal Biceps Ruptures in National Football League Players: Return to Play and Performance Analysis.

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PII: S1058-2746(20)30956-3

DOI: https://doi.org/10.1016/j.jse.2020.11.010

Reference: YMSE 5465

To appear in: *Journal of Shoulder and Elbow Surgery*

Received Date: 3 September 2020

Revised Date: 1 November 2020

Accepted Date: 8 November 2020

Please cite this article as: McGinniss A, Guinand LA, Vosbikian M, Ahmed I, Distal Biceps Ruptures in National Football League Players: Return to Play and Performance Analysis., *Journal of Shoulder and Elbow Surgery* (2021), doi: https://doi.org/10.1016/j.jse.2020.11.010.

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**Title:** Distal Biceps Ruptures in National Football League Players: Return to Play and Performance Analysis.

**Short Form/Running Title:** Distal Biceps Ruptures in NFL Players

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**Note:**

This manuscript was presented at the American Association for Hand Surgery (AAHS) Meeting in 2019, Las Vegas, NV, USA.

Ethical approval was not acquired for this study. This study is in accordance with the ethical standards of the responsible committee on human experimentation (institutional and national) and with the Helsinki Declaration of 1975, revised in 2008. This study does not contain any studies with animal subjects.

**Declaration of Conflicting Interest:**

The authors declare that there is no conflict of interest.

**Funding:**

The authors received no financial support for the research, authorship, or publication of this manuscript.

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Distal Biceps Ruptures in NFL Players

**Abstract:**

**BACKGROUND:** Distal biceps ruptures are rare injuries that tend to occur younger in high level athletic populations. There is a lack of data analyzing athletes ability to return to play, and performance, after surgery for a distal biceps rupture. **METHODS:** All National Football League players from seasons 2000-2016 that were found to have a surgically treated distal biceps rupture were included. Performance analysis and career length was conducted with a position, age, experience, and performance statistics matched control group. Data for cohort versus control, and pre-injury vs post-injury were analyzed with paired-samples Student t-test with a value of $p$ less than 0.05. **RESULTS:** Thirty-five NFL players were identified for the study and 33 were able to return to sport (94%) at an average of 351.4 (+/- 123.9) days. Offensive lineman undergoing surgery played less games per season compared to the control group $p = 0.04$). However, average seasons post surgery and post index was not found to not be significant ($p > 0.05$). Mean career length, or games per season, did not differ for post-surgical versus control group ($p > 0.05$) for all other positions. Performance scores within skill players did not prove to be significant between postoperative and control groups ($p > 0.05$). **CONCLUSIONS:** Distal biceps ruptures treated surgically in NFL players allow for return to play at a high rate. Level of performance post-surgery is similar to performance of the player pre-injury. On average NFL career length does not appear to be affected after distal biceps surgery.

**LEVEL OF EVIDENCE:** Level III; Case-Control Design; Prognosis Study

**KEYWORDS:** Distal biceps repair, National Football League, return to play, performance analysis

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Distal biceps injuries have been studied in various populations, but outcomes after surgery have not been reported in high-level athletes within the literature. Injury to the distal biceps is uncommon, at a rate around 1.2 per 100,000.[15] Typically, in athletes, distal biceps ruptures tend to affect younger populations, but are most common among 40-50-year-old men, involving the dominant extremity 86% of the time.[8] This injury does significantly affect the upper extremity through decreased elbow flexion and supination power. The mechanism of a distal biceps rupture is usually the outcome of a forceful eccentric contraction. An unexpected extension force is applied to a flexed and supinated forearm resulting in a rupture of the tendon.[8,15]

Previous studies have shown return to prior functional activity at a high level following surgical repair of a distal biceps rupture in other populations.[16] Other injuries have been studied in the National Football League players revealing the rate of return to play and performance data following injuries involving the anterior cruciate ligament, ulnar collateral ligament, forearm fractures, clavicle fractures, and tarsometatarsal joints.[5-7,13-14] To our knowledge, no data exists how elite athletes recover from distal biceps ruptures in the literature. In an epidemiological study regarding musculoskeletal injuries in the NFL, of the 4284 injuries identified during two regular seasons 2012-2013 and 2013-2014, (18%) of the injuries were upper extremity injuries.[9] Carlisle et al observed that out of all total upper extremity injuries in the NFL over a ten-year span, distal bicep ruptures accounted for at least 2% of total injuries.[3] There are a variety of positions in the NFL, and sustaining a distal biceps rupture is likely to impact certain positions differently. Certain positions, such as offensive linemen, defensive linemen, and linebackers,

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apply numerous, strenuous repetitions with their upper extremities to perform their respective positional tasks.

Our hypothesis is that NFL players who sustain a distal biceps tendon rupture treated surgically are able to return to play at a high rate, and at a level of performance comparable prior to the injury. A secondary measure in this study is to quantify the effect on the player's performance following a surgically treated biceps injury. In addition, we will compare a matched control group of NFL players to the cohort of players undergoing surgical repair.

**Methods:**

A retrospective case-control study was done, complied from National Football League players from the 2000-2016 seasons that were found to have a surgically treated distal biceps ruptures during the season or off-season were included. These players were identified on publicly available reports through media, injury reports, player stats databases, and team websites. Two separate searches were conducted by two authors of this paper. Searches were conducted on all NFL teams and players in the league during this time period. This method has been implemented in other studies that also observe the rates of return to play for other musculoskeletal injuries in athletes, including Jack RA et al and Sochacki KR et al.[5-7,11,13,14]

Each of the players identified was able to have data collected to formulate a return to play (RTP) cohort. Return to play was defined as having played one regular season NFL game after sustaining a distal biceps rupture that was treated surgically. A player was considered to not return to play if they did not participate in at least one regular season game following surgery.

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Inclusion criteria were all players on an active NFL roster from 2000-2016 who sustained a distal biceps rupture treated surgically. Players were included in the study if the two separate searches independently found sources confirming a player sustained the injury and underwent surgery. Any players that had conflicting reports were excluded from the data collection. Players injured in the 2017-2018 season were excluded as they did not have a full year to allow for return to play.

Performance analysis was conducted with a position, age, experience, and performance statistics matched control group. Players stratified by their respective position including cornerback "CB," defensive end "DE," defensive tackle "DT," linebacker "LB," tight end "TE," running back "RB," safety "S," and offensive linemen "OL". One matched control was selected for each player in the cohort group within ± one year of age, experience, and performance score. Data was collected and analyzed for pre-injury and post-injury performance metric within the cohort group with a paired-sample t-test with value $p<0.05$. Case and control performances scores were calculated using a previously published standardized scoring system that accounts for variables that are specific to an offensive or defensive player.[7] Players in the control group were given an index date to match the injury date of the cohort group. Data for cohort versus control and pre-injury vs post-injury were analyzed with independent-samples t-test with $p<0.05$. SPSS version 25 (IBM, Armonk, NY, USA) was used for statistical analysis.

**Results:**

Thirty-five NFL players were identified with distal biceps repairs during the study period. There were no reports of conflicting data, therefore all players were included in the cohort. Most players identified were offensive linemen n= 12 (35%). The average age of players undergoing

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distal biceps repair was 28.5 ± 3.5 years old. One player ruptured his contralateral biceps the following season after undergoing surgery. He was counted as two separate data points. Thirty-three players, (94%), were able to return to sport after undergoing distal biceps surgery. The average return to play (RTP) measured in days was 324.7 ± 112 for the entire cohort (Table I). The distribution of the RTP in days by player position is demonstrated in Figure I. The linebacker position had one player who did not play the following season in a regular season game. He was on a roster, but since he didn't play in a game until 2 years post injury, this increased the average for time to return to play for the LB position. The two players (6%) unable to return to sport were both offensive linemen, and further along in their career at 9 and 14 years, respectively. The average amount of seasons played before sustaining a distal biceps injury in the defensive cohort was 8.3 ± 3.3 seasons and 9.6 ± 2.4 for the offensive player cohort.

No players returned the same season after having a surgical repair performed. There was found to be no significant difference in player demographics between the cohort and control group in defensive players, Age ($p$=1.0), weight ($p$=.84), height ($p$=.44), BMI ($p$=.34) (Table II). There was also no significance in pre and post-surgery cohort versus pre and post-surgery control performance scores for each position in both defensive and offensive players (Table III). There was a significant difference in DT performance scores found in the cohort group pre and post surgery, 2.9 ± 0.7 versus 1.7 ± 0.3 ($p$ = 0.02). When the DT cohort was compared to controls, there was no significant difference in performance scores post surgery. Only two offensive players were included in the study, one RB and one TE. Therefore, we were unable to assess performance score significance pre and post-surgery for these positions. Offensive linemen in the post surgery cohort played an average of 11.1 ± 3.4 games per season versus 13.6 ± 2.1 games per season by the control group post index ($p$ = 0.04) (Table V). The career length in

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seasons, post surgery and post index, did not differ between the two OL groups ($p$ = 0.79). (Table IV). All other positions did not have any significant difference for games played or season played, post surgery versus post index.

**Discussion:**

Distal bicep ruptures in NFL players were shown to cause a prolonged absence from participation in training and game time. Despite the impact of the injury, there is evidence supported by the literature indicating above average results and return to prior function when opting for surgical treatment.[17] Elbow flexion, and supination strength, as well as endurance returned as expected.[1,2,12] It was noted in two studies, involving an athletic and military population, that the results were independent of the surgeon's preference of using a single-incision or 2-incision technique, and best when surgical repair occurred within two weeks of the injury.[1,17] It should be emphasized that these recommendations cannot be assessed from this study, and could be a subject addressed in a future analysis with a larger sample of NFL players.

The goals of this study were to determine the RTP rates in NFL players after distal biceps repair and establish any significant trends in pre surgery versus post surgery performance, as well as performance compared to a control group of players (Table III). The control group was matched by respective position, age, years of experience, and post index performance scores. Also, the study sought to determine if there were any significant changes in games per season, and career length compared to their matched control post surgery (Table IV). The study's standing hypothesis was confirmed with a high RTP rate of (94%) and similar career lengths and games per season as their matched controls. The exception being offensive linemen that had a significant decrease in games per season post surgery compared to their control ($p$ =.04).

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However, the OL career lengths per season post surgery were not affected when compared to the control group.

The offensive linemen composed the largest group of players that had a distal biceps surgery with n=14 (40%), two of which retired and did not return for the following season. We hypothesize that the disproportionate amount of distal biceps injuries is due to the position's unique role in the game. The offensive linemen (OL) tend to engage defensive linemen in a way in which it puts substantial stress on the biceps. This sort of activity predisposes the OL position to greater forceful eccentric contractions of the biceps which could lead to rupture. The predilection of the offensive and defensive linemen to injury was also noted by Carlisle et al in an epidemiologic study of upper extremity injuries in the NFL.[3,10] The two players in the cohort that did not return to play after sustaining distal biceps ruptures occurred after playing 9 and 14 seasons. On average the cohort sustained distal bicep ruptures after 8-9 seasons. This may be a clue that the injury is more predisposed to players after more repetitive strain is placed on the tendon due to longer careers in the NFL.

The strengths of this study are in the strict inclusion and exclusion criteria when formulating a cohort and control group, in combination with the thorough literature review that was performed on the topic. Another strength is this study provides a general timeline of when football players can expect to return to play following surgery for a distal bicep rupture. The data within this study can also help physicians educate athletes on the expectations of performance following surgery. There were also some limitations present as well. The low incidence of the injury in itself, may have led this study to be underpowered to show a difference in the results. One is the inability to access data about players with distal bicep tears, along with surgical method used for treatment directly from the NFL. The use of public available data can

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have selection, reporting, and observer bias .To overcome this limitation the confirmation of both diagnosis and surgical treatment had to be independently confirmed by at least two sources. Others included the lack of literature regarding distal bicep tears in high-level athletes. While there was literature regarding distal bicep tears in athlete populations, there were not many comparable studies regarding distal bicep tears in NFL athletes. The final limitation was the lack of performance variables to calculate a performance score for offensive linemen. Future considerations for would be to observe the impact on RTP and performance when using 1-incision vs. 2-incision technique in NFL athletes. Also, the same study can be applied to other high-level athletes that endure unique stresses on their biceps such as rugby players, bodybuilders, or Olympic weightlifters.

**Conclusion:**

Out of the thirty-five players with distal biceps tear that were confirmed, thirty-three were able to return to a high level of competition indicated by RTP rates. Spanning all positions, the average time to return to play was 324.7±112.2 days, or about a year. Additionally, the outcome of this study provides information that NFL players that undergoing surgical repair for a distal biceps tears did not have a significant effect on their performance when returning for at least one season post-surgery. In general, performance scores after surgical repair were comparable to those prior to surgery with the exception of defensive tackles, where there was a significant change in performance scores in the cohort ($p < .05$). When comparing performance scores to their matched controls, there were no significant findings. Given the prolonged healing periods for players that underwent distal bicep repair surgically and high rates of RTP, this study can be of service to further educate players, coaches, and related staff on the impact of the injury.

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Figure I. Average Return to Play (days) by player position and average for cohort. CB = cornerback; DE = defensive end; DT = defensive tackle; LB = linebacker; S = safety; RB = running back; OL = offensive line; TE = tight end.

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**Table I. Numbers of Distal Bicep Repairs With Return to Play (RTP) by Position[a]**

| Position | n | RTP, n | RTP, % | RTP (Days) |
|---|---|---|---|---|
| CB | 1 | 1 | 100.0 | 314.0 |
| DE | 4 | 4 | 100.0 | 213.5 ± 68.0 |
| DT | 5 | 5 | 100.0 | 359.4 ± 41.7 |
| LB | 3 | 3 | 100.0 | 492.7 ± 221.9 |
| S | 6 | 6 | 100.0 | 349.8 ± 74.8 |
| RB | 1 | 1 | 100.0 | 250.0 |
| TE | 1 | 1 | 100.0 | 462.0 |
| OL | 14 | 12 | 85.7 | 288.3 ± 80.4 |
| Total | 35 | 33 | 94.0 | 324.7 ± 112.2 |

[a]CB = cornerback; DE = defensive
RB = runningback; TE = tight end;

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**Table II. Cohort and Control Group Demographic Data[a]**

| Demographics | Cohort | Control | P value |
|---|---|---|---|
| Defensive NFL players | | | |
| NFL seasons played (yrs) | 8.3 ± 3.3 | 8.29 ± 3.1 | (.92) |
| Age (yrs) | 28.0 ± 3.3 | 27.9 ± 3.1 | (1.00) |
| Weight (kg) | 114.1 ± 27.2 | 112.42 ± 22.3 | (.84) |
| Height (m) | 1.85 ± 0.1 | 1.87 ± 0.1 | (.44) |
| BMI | 32.26 ± 6.6 | 30.5 ± 4.9 | (.34) |
| Games Postsurgery | 23.2 ± 18.4 | 25.3 ± 22.0 | (.75) |
| CB/DE/DT/S/LB | 1/4/5/6/3 | 1/4/5/6/3 | |
| Offensive NFL players | | | |
| NFL seasons played (yrs) | 9.6 ± 2.4 | 13.0 ± 2.1 | |
| Age (yrs) | 28.5 ± 3.3 | 28.0 ± 5.7 | |
| Weight (kg) | 135.6 ± 13.6 | 115.5 ± 0.7 | |
| Height (m) | 1.9 ± 0.5 | 1.90 ± 0.1 | |
| BMI | 36.1 ± 3.0 | 32.0 ± 2.8 | |
| Games Postsurgery | 33.5 ± 28.0 | 79.0 ± 58.0 | |
| RB/TE/OL | 1/1/12 | 1/1 | |

[a]Output values are reported as means. CB = cornerback; DE = defensive end; DT = defensive tackle; S = safety; LB = linebacker; RB = runningback; TE = tight end.

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**Table III.**

**Pre- and Postsurgery Performance Scores by Position for Cases Pre- and Postindex Performance Scores for Controls[a]**

| | | Cohort | | | Control | | | |
|---|---|---|---|---|---|---|---|---|
| Position | N | Presurgery | Postsurgery | P[b] | Preindex | P[c] | Postindex | P[d] |
| CB | 1 | 5.8 | 4.8 | | 4.2 | | 3.2 | |
| DE | 4 | 3.0 ± 0.8 | 2.9 ± 1.4 | .87 | 2.8 ± 1.8 | .87 | 2.6 ± 0.2 | .65 |
| DT | 5 | 2.9 ± 0.7 | 1.7 ± 0.3 | *.02 | 2.2 ± 0.8 | .22 | 3.2 ± 2.5 | .25 |
| LB | 3 | 4.2 ± 3.3 | 4.4 ± 3.4 | .83 | 6.8 ± 1.5 | .30 | 5.4 ± 1.5 | .67 |
| S | 6 | 4.8 ± 2.3 | 3.2 ± 2.3 | .16 | 3.3 ± 1.5 | .19 | 7.1 ± 7.5 | .27 |
| RB | 1 | 3.2 | 4.8 | | 3.2 | | 1.9 | |
| TE | 1 | 1.9 | 0.0 | | 1.9 | | 4.4 | |

[a]CB = cornerback; DE = defensive end; DT = defensive tackle; S = safety; LB = linebacker; RB = runningback; TE = tight end; OL = Offensive Line.
[b]Student *t* test results for pre- vs. postsurgery performance scores
[c]Student *t* test results for presurgery vs. preindex performance scores
[d]Student *t* test results for postsurgery vs. postindex performance scores
*Statistically significant difference ($P < 0.05$)

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**Table IV. NFL Career Length Postsurgery and Postindex[a]**

| | | Career Length (yrs ± SD) | | |
|---|---|---|---|---|
| Position | N | Cohort | Control | P Value |
| CB | 1 | 3.0 | 3.0 | |
| DE | 4 | 2.0 ± 1.4 | 2.5 ± 1.3 | .62 |
| DT | 5 | 1.4 ± 0.5 | 1.8 ± 1.1 | .49 |
| LB | 3 | 1.0 ± 0.0 | 1.0 ± 0.0 | |
| S | 6 | 1.8 ± 1.6 | 2.2 ± 1.9 | .75 |
| TE | 1 | 1.0 | 1.0 | |
| RB | 1 | 3.0 | 3.0 | |
| OL | 12 | 2.8 ±1.9 | 2.6 ± 1.9 | .79 |
| Overall | 33 | 2.2 ± 1.6 | 2.2 ± 1.6 | .94 |

[a]CB = cornerback; DE = defensive end; DT = defensive tackle; S = safety; LB = linebacker; RB = runningback; TE = tight end; OL = Offensive Line.
*Statistically significant difference (P < 0.05)

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**Table V.**

**Pre- and Postsurgery Games per Season by Position for Cases Postindex Games per Season for Controls[a]**

| | | Games per Season (mean ± SD) | | | | |
|---|---|---|---|---|---|---|
| | | Cohort | | | Control | |
| Position | N | Presurgery | Postsurgery | P[b] | Postindex | P[c] |
| CB | 1 | 11.0 | 13.0 | | 13.0 | |
| DE | 4 | 10.6 ± 5.4 | 11.8 ± 3.8 | .71 | 11.7 ± 5.2 | .97 |
| DT | 5 | 11.6 ± 3.2 | 13.9 ± 2.1 | .27 | 13.2 ± 3.3 | .70 |
| S | 6 | 12.9 ± 3.3 | 9.1 ± 5.7 | *.05 | 11.7 ± 4.8 | .42 |
| LB | 3 | 14.2 ± 1.4 | 14.3 ± 2.1 | .92 | 9.7 ± 4.7 | .22 |
| TE | 1 | 10.0 | 14.0 | | 15.0 | |
| RB | 1 | 12.0 | 3.0 | | 12.7 | |
| OL | 12 | 11.0 | 11.1 ± 3.4 | *.05 | 13.6 ± 2.1 | *.04 |

[a]CB = cornerback; DE = defensive end; DT = defensive tackle; S = safety; LB = linebacker; RB = runningback; TE = tight end; OL = Offensive Line.
[b]Student *t* test results for pre- vs. postsurgery games per season
[c]Student *t* test results for postsurgery vs. postindex games per season
*Statistically significant difference (P < 0.05)

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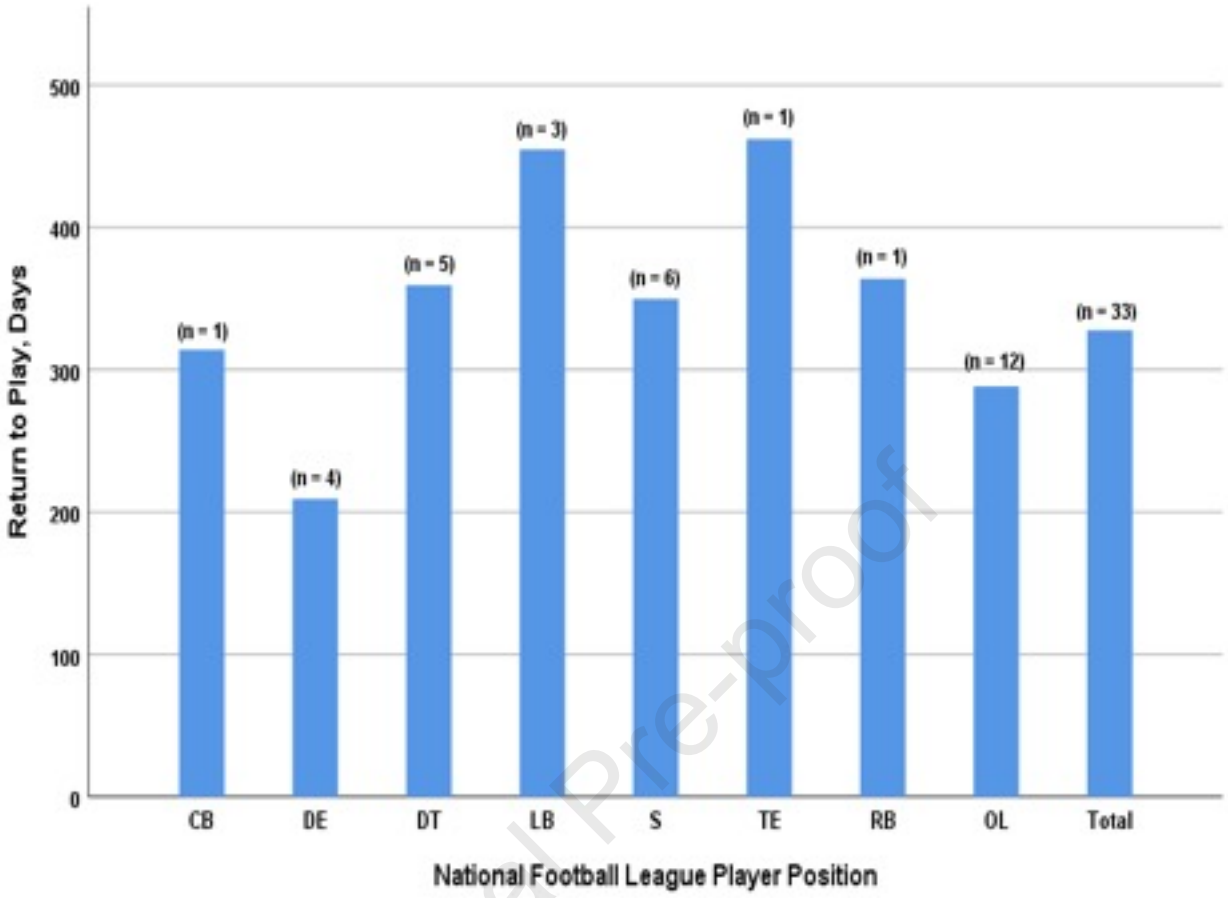


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